How Much Does Time Matter?
A Comparative Study of Aggregation, Temporal Patterns, Dynamics and Community Evolution in Real-World Temporal Networks

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Abstract

Real networks are made of interactions that happen at specific times, yet they are routinely analysed as static aggregated graphs. We ask how much this discards. We organise the temporal-network literature around four questions—representation, activity patterns, structural evolution and function—distil seven testable hypotheses (H1–H7), and define a common pipeline of window analysis, randomized reference models and dynamic-community tracking on three networks of different type and time scale: face-to-face contacts in a primary school, online messaging among students, and a computer-science citation network. The planned cross-network comparison uses normalised time effects *rather than raw metrics. The completed citation-network study contains 19,150 papers and 73,269 strictly older-target citations. It supports sublinear degree preference and age-dependent citation choice, with a densification exponent of 1.343, but no detected shrinkage of the sampled effective diameter. A three-parameter citation model fits selected histories well without an established MAE advantage over linear extrapolation. Contact and messaging experiments remain pending, so a cross-network ranking of temporal sensitivity is not yet inferred.*

1. Introduction

Real networks evolve: nodes and links appear and disappear, and interactions take place at specific moments—a conversation lasts a minute, a message is sent at a given second, a paper cites earlier work in a given year. Classical network analysis nevertheless collapses all of this into one static graph in which two nodes are either linked or not. Such aggregation is convenient, but it discards when links are active, how often they recur, in which order they occur and how the network changes. Unlike for static graphs, connectedness is not transitive in time, so a temporal network cannot be reduced to a simple graph without loss [10]. The temporal-network literature [10, 11, 19] has shown that each of these ingredients can change the conclusions drawn about structure, communities, and processes such as epidemics.

Most of this evidence, however, comes from a single dataset or a single method. It is therefore hard to tell *how much* time matters in general, and for which kinds of systems. We address two research questions. **RQ1:** how does temporal information change our understanding of the structure and function of real networks, and how does this effect differ across network types? **RQ2:** which hypotheses from the temporal-network literature (H1–H7, Tab. 2) hold in which systems?

Approach. The report is organised around four questions that recur in every section (Fig. 1): (I) *representation*—what is lost by aggregation and by the choice of time window? (II) *activity patterns*—are interactions bursty, repeated and persistent? (III) *structural evolution*—how do networks grow, age and reorganise? (IV) *function*—does time ordering change reachability, centrality and spreading? Each question is turned into a literature hypothesis, a metric and a randomized reference model [5], and applied to three networks that differ in edge meaning, generating mechanism and time scale (seconds, seconds, years): face-to-face contacts, online messages and citations. Because raw metrics are not comparable across such systems, datasets are compared via *normalised time effects* (Sec. 4.7). We do not propose new models; the contribution lies in the common framework and the comparison.

Contributions. (1) A question-driven synthesis of how network scientists study temporal networks; (2) a common, reproducible pipeline that applies it to three types of real data; (3) a cross-network verdict on H1–H7, stating where time information is essential and where a static view suffices.

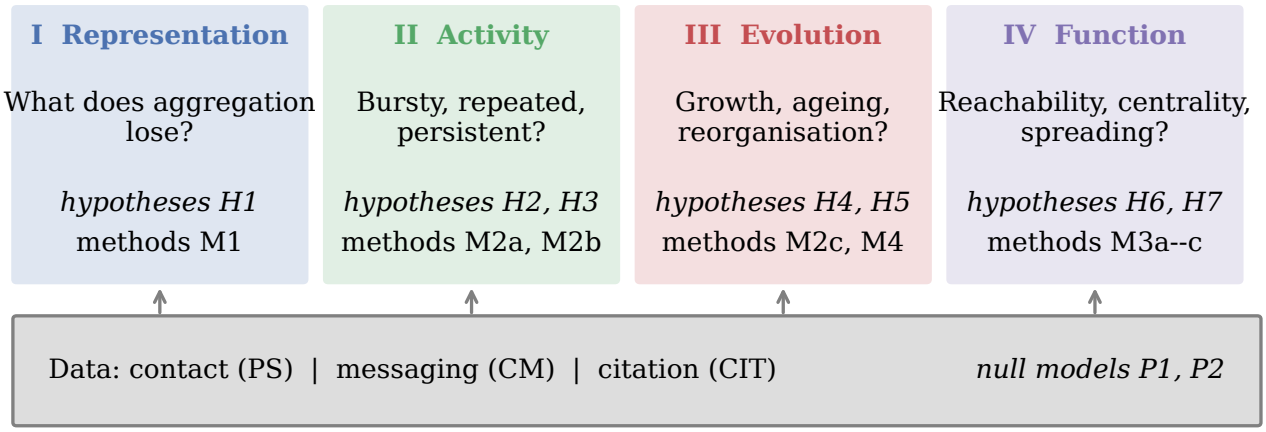


Figure 1. Structure of the study. Four questions (Parts I–IV) map to seven hypotheses and four method groups; the same order is used in Related Work, Methods, Results and Discussion. All analyses use the same three datasets and the same randomized reference models.

Roadmap. Section 2 reviews related work, Sec. 3 describes the data, Sec. 4 the methods, Sec. 5 the results, Sec. 6 discusses them and Sec. 7 concludes.

2. Related Work

We organise the literature by question rather than paper by paper; the subsections follow Parts I–IV of the Introduction. General frameworks and reviews are given by Holme [10], Holme and Saramäki [11] and the book of Masuda and Lambiotte [19]. A central point of these reviews is that the basic unit is a *contact* (two nodes and a time), and that connectedness is no longer transitive: if a meets b and later b meets c, a path from a to c exists, but not the other way round. Hence a temporal network cannot be turned into a simple graph without losing information or changing what a node means, and many static-network methods need nontrivial generalisations [10].

2.1. Representing time: aggregation, snapshots and windows

Temporal networks can be stored as contact sequences, interval graphs or sequences of snapshots [11]. Lossy representations such as weighted aggregates or reachability graphs reduce the data to a static object; lossless ones keep all contacts [10]. Each choice involves a trade-off: snapshots make static tools applicable but require a window length, whereas the full event sequence keeps all information. The effect of this choice has been quantified for communication data: window size and placement change the structure of the aggregated network [16], and coarser temporal resolution distorts properties of the time-varying network in a way that depends on the system [27]. This motivates **H1**: aggregation and window choice change structural conclusions.

2.2. Temporal activity patterns

Human activity is bursty: inter-event times are heavy-tailed rather than exponential [1], and consecutive intervals are correlated, which Goh and Barabási [7] capture with a burstiness coefficient and a memory coefficient. Social ties are also repeated and persistent over long periods [30], while the set of active links turns over. Temporal motifs [15, 25] and activity-driven models [26] describe and generate such patterns. This gives **H2** (activity is bursty and has memory) and **H3** (ties are repeated and persistent while the network turns over).

2.3. Structural evolution: growth, ageing and communities

Growth is classically explained by preferential attachment [2], whose strength can be measured directly from growth data [12, 22]. Real systems usually show ageing, where old nodes attract fewer new links [4], and in citations this is captured by a relevance decay [35]. Growing networks also densify, with the number of edges growing superlinearly in the number of nodes [17, 18]; node and link deletion alter these laws [20]. This gives **H4**: growth follows preferential attachment with ageing, and networks densify.

At the mesoscale, multilayer modularity couples partitions across time [21], communities can be tracked across snapshots through matching [8] and change can be mapped between partitions [29]; Rossetti and Cazabet [28] survey the field. Per-snapshot partitions are typically obtained with Louvain [3] or Leiden [34]. This gives **H5**: communities reorganise over time and a static partition hides this.

2.4. Time and network function

Because interactions occur in sequence, only time-respecting paths can carry information: reachability in contact sequences is much lower than in the aggregated graph [9], and path lengths and centrality change accordingly [14, 23]. Burstiness and temporal correlations further slow down spreading even when the static topology is unchanged [13]. These results give **H6** (time ordering restricts reachability and changes centrality rankings) and **H7** (burstiness and temporal correlations slow down spreading).

2.5. Randomized reference models

Randomization is more versatile for temporal than for static networks, where it is essentially limited to degree-preserving link shuffling, because one can switch off individual kinds of correlation [10]. Randomized reference models act as hypothesis tests: each one preserves some properties of the data and destroys others, so differences between data and model reveal which feature matters [5]. We use them in Parts II–IV (Sec. 4.6).

Time scales and network types. Temporal networks are usually defined as systems whose evolution is not much slower than the processes running on them, and a single system contains many time scales (seconds, hours, days) [10]. Contact and communication data are event sequences with such a hierarchy, whereas citation networks are acyclic by construction and all out-links of a paper appear at one instant [10]. This is why we expect, and test, different sensitivity to time across network types.

Gap. Most studies examine one dataset or one method. Few compare, within a single framework, how sensitive different *types* of networks are to temporal information. The hypotheses distilled above (Tab. 2) allow such a comparison.

3. Data

We use one network per type, chosen to differ in edge meaning, generating mechanism and time scale (seconds, seconds, years); data cards are in Tab. 1 and activity time series in Fig. 2.

Face-to-face contact. The primary-school data [6, 32] were collected by the SocioPatterns collaboration with wearable proximity sensors over two school days. A contact is recorded when two people are face to face within about 1.5 m for at least 20 s. Each individual carries a class label, which we use as ground truth for communities. The data are released under CC BY-NC-SA.

Type	Dataset	N	Events	Span	Res.
Contact	Primary school (PS)	242	125 773	2 d	20 s
Comm.	CollegeMsg (CM)	1 899	59 835	193 d	1 s
Citation	DBLP–DB (CIT)	19 150	73 269	50 yr	1 yr

Table 1. Data cards. N: nodes; Res.: temporal resolution. CIT spans publication years 1970–2020, with primary analysis in 1990–2020. PS: undirected, class labels, CC BY-NC-SA; CM: directed, no labels; CIT: directed, acyclic.

Online communication. CollegeMsg [24] contains time-stamped private messages between 1 899 users of an online student community over 193 days at 1 s resolution. Edges are directed, and there are no node labels.

Scientific publications. We use AMiner DBLP-Citation V14 [33], selected by a fixed set of database venue families and their declared name variants. Streaming 5,259,858 records yields 19,150 publications from 1970–2020 and 73,269 distinct citations to strictly older selected papers. These are within-subgraph citations, not global citation counts; 6,709 nodes are isolates. We save 5,486 same-year references separately, exclude 6,390 later-year references and 494 self-loops, and remove 37,440 repeated reference entries. The graph is acyclic under the strict year rule. AMiner specifies research use; no additional open-data licence is inferred. The recent fall in observed references despite growing paper counts motivates a cutoff-2015 sensitivity check and cautions about coverage and censoring.

Preprocessing. All data are converted to a common event format (u, v, t) with integer time stamps (seconds or years) plus a table of node labels. Self-loops and exact duplicate events are removed, and the first and last partial windows are trimmed. Each source has known limits: citation data have year-level resolution, the contact data cover only two days, and the sensors sample only participants who wear them. The three datasets differ in edge meaning (proximity, message, citation), directedness and time scale by design, so that agreement across them is not an artefact of one data type.

4. Methods

Every analysis follows the same template: hypothesis from the literature → metric → randomized reference model. Table 2 lists the seven hypotheses, the methods (M1–M4) that test them and the datasets where each method is meaningful (19 method–dataset pairs). The empty cells are deliberate: CIT is a year-resolution DAG without fine-grained events (no M2a, M2b, M3) and without community reorganisation (no M4), while PS and CM have no growth process (no M2c).

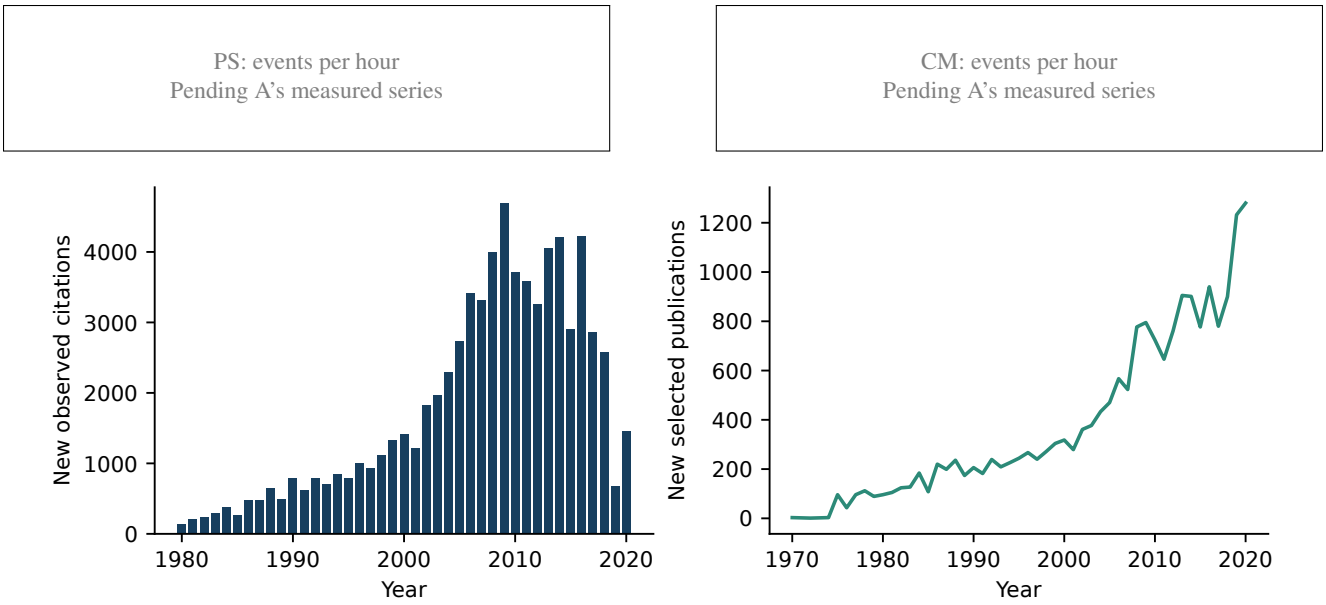


Figure 2. Activity overview. The lower panels are measured CIT annual citations and selected publications; upper panels reserve the pending PS/CM series. CIT's late reference decline alongside rising publication counts motivates coverage and observation-cutoff checks.

Part	ID	Hypothesis	Source	Method	Null	PS	CM	CIT
I	H1	Aggregation and window choice change structural conclusions	[16, 27]	M1	–	✓	✓	✓
II	H2	Activity is bursty and has memory	[7]	M2a	P1	✓	✓	
	H3	Ties are repeated and persistent; the network turns over	[30]	M2b	P1	✓	✓	
III	H4	Growth follows preferential attachment with ageing; networks densify	[12, 17, 35]	M2c-1–4	P3			✓
	H5	Communities reorganise over time; the static partition hides this	[8, 21]	M4	P1	✓	✓	
IV	H6	Time ordering restricts reachability and changes centrality rankings	[9, 14, 23]	M3a, M3b	P1, P2	✓	✓	
	H7	Burstiness and temporal correlations slow down spreading	[13]	M3c	P1, P2	✓	✓	

Table 2. Hypotheses tested in this work, with methods, null models (Sec. 4.6) and applicable datasets.

4.1. Notation

Let $\mathcal{E} = \{(u, v, t)\}$ be the event sequence. The static aggregate G contains a link (u, v) if at least one event occurs between u and v. A snapshot $G_{\Delta t}(k)$ contains the events in $[k\Delta t, (k+1)\Delta t)$, and the cumulative network $G(\leq t)$ contains all events up to t.

4.2. Aggregation and time windows (M1)

For each window length Δt on a grid (PS: 5 min to 1 day; CM: 1 h to 1 month) we compute density, mean degree, clustering coefficient and giant-component size of the snapshots and of the aggregate. For CIT, yearly and cumulative networks are compared. *Window sensitivity* is the relative change of a metric across the grid. Rankings are handled separately in M3b. Figure 3 shows why aggregation can mislead: a path present in the static graph may not be traversable in time.

4.3. Activity patterns (M2a, M2b)

M2a measures the inter-event time distribution of each node pair and of each node, fitted with a heavy-tailed model, together with burstiness $B = (\sigma-\mu)/(\sigma+\mu)$ and memory M, the correlation of consecutive intervals [7]. **M2b** measures

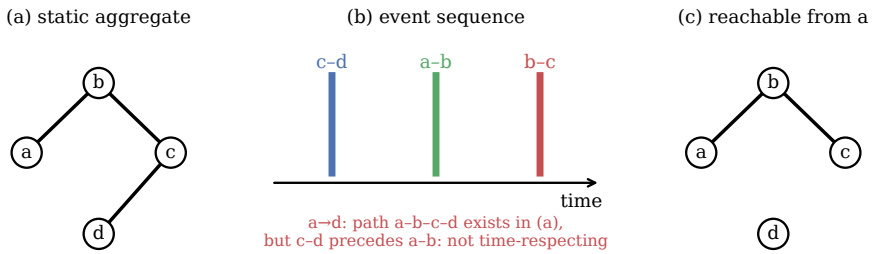


Figure 3. Aggregation adds paths that do not exist in time. In (a) a appears to reach d; in (b) the link c–d is active before a–b, so in (c) only b and c are reachable from a.

the edge repetition rate, tie lifetime and the Jaccard similarity of consecutive snapshots, which quantifies the turnover of nodes and links.

4.4. Structural evolution (M2c, M4)

Citation data and time resolution. We stream AMiner DBLP-Citation V14 [33] and retain an induced subgraph of database venues selected by a fixed whitelist (SIGMOD, VLDB/PVLDB, ICDE, PODS, EDBT, ICDT, TODS, VLDB Journal, Data & Knowledge Engineering, and Information Systems). All selected papers published in 1960–2020 are retained, including isolates. A distinct citation $u \to v$ occurs in the publication year of u. The primary analysis requires $y_v < y_u$; same-year citations are saved for structural sensitivity analysis because their order cannot be resolved. We exclude self-citations to the same paper, duplicate edges and future references. Author self-citations are not removed. Counts refer to citations within the selected venues, rather than global citation impact. The primary analysis years are 1990–2020, initialized using the available earlier history.

M1: annual and cumulative citation structure. Annual and cumulative networks use the same node denominator: all selected papers already published by year t. Directed density is $E/[N(N-1)]$ and mean total degree is $2E/N$. Clustering, component counts and largest-component fractions use the weak undirected projection, including isolates. Trailing windows of 1, 2, 5 and 10 years provide a window-sensitivity check; sensitivity is $(\max - \min)/\mathrm{mean}$ over the grid at each endpoint, then averaged over endpoints.

M2c-1: preferential attachment. Following the degree-class approach of Jeong et al. [12], we freeze in-degree at the start of each citing year and count eligible nodes n_{tk} and new citations h_{tk} in each prior-degree class k. We estimate α by conditional likelihood with probabilities proportional to $n_{tk}(k+1)^{\alpha}$, conditioning on the total number of citations in each year. The offset admits citations to previously uncited papers; offsets 0.5 and 2 provide sensitivity checks. Confidence intervals use 400 resamples of citing-year blocks. These are descriptive uncertainty intervals; neither positive α nor its departure from an age-only null identifies a causal mechanism.

M2c-2: ageing. For ages 1–60, the target probability in citing year t is proportional to $n_{ta}f(a)$, where n_{ta} counts available papers of age a. A conditional target likelihood compares discrete-grid lognormal, exponential, power-law and uniform kernels using AIC on identical observations and support. We repeat the comparison by publication cohort (1970–1989, 1990–1999, 2000–2009), accounting for each cohort's actual exposure. This estimates marginal ageing; degree, fitness and venue coverage can also influence this curve.

M2c-3: densification and diameter. We fit $\log E(t) = a \log N(t) + b$ over 1990–2020 and repeat over split periods and an earlier observation cutoff. Year-resampling intervals are descriptive because cumulative observations are dependent. Effective diameter is the 90th percentile of weak shortest-path distances within the largest connected component, using 128 uniformly sampled source nodes per year. Sensitivity checks use 64 and 256 sources. We save the component size and number of sources with every estimate [17].

M2c-4: citation dynamics and prediction. We fit the Wang–Song–Barabási curve [35],

$$c(t) = m\left[\exp\left\{\lambda\Phi\left(\frac{\log t - \mu}{\sigma}\right)\right\} - 1\right], \qquad (1)$$

where m is fixed to the mean observed within-subgraph reference count of papers published through 1989. Bounded multistart least squares minimizes squared $\log(1+c)$ errors; parameter bounds and boundary flags are preserved. For descriptive full-history fits we randomly select at most 500 papers born in 1990–2005 with at least 10 citations by age 15

and fit years 1–15. The resulting collapse is in-sample evidence only. For prediction, selection uses only the first five years: at least five observed citations by age five, followed by a fixed random sample of at most 500 papers. Using only years 1–5, we compare WSB against no-ageing PA, $c(t) = m[\exp(rt) - 1]$, and linear extrapolation, $c(5)t/5$. All models predict the same papers at age 15 and at the observed endpoint in 2020; these are censored observed totals, not unobserved lifetime citations. We report MAE, median absolute error, RMSLE, paired bootstrap differences (1000 resamples) and an interior-fit sensitivity check. Boundary fits remain in the primary evaluation so that difficult cases are not silently discarded.

P3: citation-target randomization. Twenty realizations per model preserve each paper's publication year and exact number of distinct observed references, with no replacement and strictly older targets. P3a selects eligible targets uniformly. P3b weights targets by a nonparametric age kernel fitted by conditional target likelihood, accounting for each citing year's available age distribution, and removes degree preference. Its age weights are calibrated from the data, so agreement on ageing is not an independent confirmation. We report null means and sample standard deviations; empirical tail probabilities have minimum resolution $1/21$. Both models fix $N(t)$ and $E(t)$ and hence exactly preserve the densification exponent. They can test diameter and target preference, but cannot supply a significance test for densification itself.

Reproducibility. All significant CPU workloads run on Vanda compute nodes through PBS without GPU allocation. Fixed seeds, input/source checksums, environment lock file, filtering counts, individual forecasts and every randomized edge list are saved. Robustness checks cover time splits, cutoff 2015, degree offsets, same-year edges, core database venues, diameter sampling, and 10 versus 20 null realizations.

M4 detects communities in every snapshot (Leiden or Louvain), matches them across snapshots [8] and reports modularity and the number of communities over time, the normalised mutual information (NMI) between consecutive snapshots, the NMI between the dynamic and the static partition, and for PS the NMI with the class labels. The resolution parameter is fixed and its effect is checked.

4.5. Paths, centrality and spreading (M3)

M3a computes earliest-arrival time-respecting paths and the temporal reachability ratio R (fraction of node pairs connected by a time-respecting path, relative to the static aggregate). **M3b** compares static and temporal closeness rankings by Kendall's τ. **M3c** runs an event-driven SIR process on the event sequence with at least 100 random seeds; SI with certain transmission is equivalent to M3a, so only SIR is run.

4.6. Randomized reference models

No single temporal counterpart of the configuration model exists, so one switches off selected correlations with different models and compares the outcomes [5, 11]. For PS and CM we use two complementary models, which correspond to *randomly permuted times* (RP) and *randomized edges* (RE) in the naming of Holme and Saramäki [11]. **P1** (RP) permutes timestamps among events: it keeps the static graph and the number of contacts of every pair, and destroys burstiness and temporal correlations. **P2** (RE) rewires the static graph in a degree-preserving way while each pair's contact sequence moves with its link: it destroys the wiring diagram but keeps per-link burstiness and the overall event rate (Fig. 4). Comparing the two separates the effect of time from that of structure. Combined models (RE+RP) and fully random times (RT) are possible extensions. Each is run with at least 20 realisations and reported as mean $\pm$ standard deviation.

4.7. Measuring the effect of time

Raw metrics cannot be compared across a contact network, a message network and a citation network, so we define one normalised measure per hypothesis: H1 window sensitivity (relative change of a metric over the Δt grid); H2 $B_{\mathrm{real}} - B_{\mathrm{null}}$; H3 mean consecutive-snapshot Jaccard similarity; H4 exponents α and a; H5 mean consecutive-snapshot NMI and dynamic-vs-static NMI; H6 reachability ratio R and Kendall τ(static, temporal); H7 slowdown $T_{\mathrm{real}}/T_{\mathrm{null}}$, where T is the time to the epidemic peak.

4.8. Implementation

All code is in Python. We use NetworkX for static graph analysis, Raphtory [31] for time-respecting paths, tnetwork for dynamic communities, and Leiden/Louvain for per-snapshot partitions. Null models, the attachment-kernel estimator and the SIR process are custom code, unit-tested on hand-built examples whose answers are known (reachability, SIR, preserved quantities of P1 and P2). Random seeds are fixed and every figure is generated by one script.

5. Results

The results follow the four parts of Sec. 4. Each subsection presents the applicable datasets and their available evidence; Sec. 5.6 collects the verdicts. The CIT results below are measured outputs of all five B experiments, including 40 citation-target null graphs. Blue text elsewhere denotes untested expectations for PS/CM, and grey boxes denote their pending figures. These draft expectations are not evidence in the measured summary.

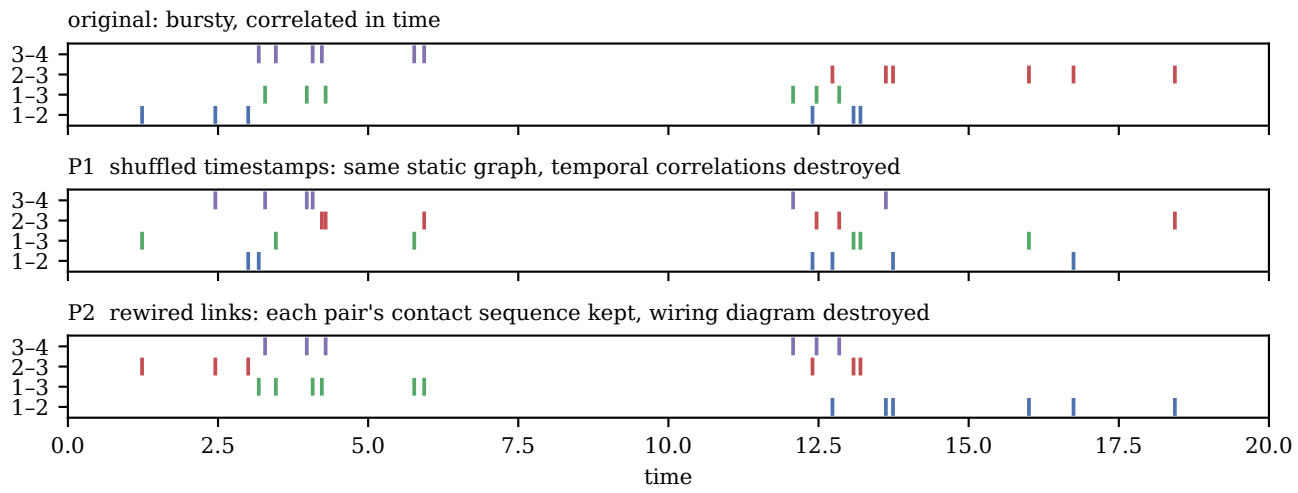


Figure 4. Toy illustration of the randomized reference models. Rows are node pairs, ticks are events. P1 permutes the timestamps, P2 permutes which pair each event belongs to.

5.1. Aggregation and time windows

Contact and messaging networks (PS, CM). *[Outline] Which conclusions are stable under window choice, and over which Δt range?*

[Expected] Density and clustering of the snapshots rise quickly with Δt and approach the aggregate only for windows comparable to the whole observation period. The giant component of short-window snapshots is small, and percolates at a window that is much shorter for CM than for PS because PS has a daily rhythm (lessons, breaks). Conclusions about "the" clustering or density of the network therefore depend strongly on Δt.

Citation network (CIT). In 2020, annual and cumulative graphs on the same 19,150 born nodes contain 1,450 and 73,269 citations. Mean total degree is 0.151 versus 7.652, weak largest-component fraction 0.052 versus 0.641, and cumulative weak clustering 0.124. Annual clustering is identically zero: strict older-target citations make each annual snapshot bipartite. This is a consequence of the representation, not independent evidence for temporal correlations. Over 1-, 2-, 5- and 10-year windows, relative ranges average 1.798 for density/degree, 2.361 for clustering and 1.114 for largest-component fraction. **CIT verdict on H1:** window choice changes structural descriptions; some differences are definitional. The node denominator and year convention must accompany every comparison.

Window interpretation. Figure 7 shows that longer windows connect a larger fraction of born papers and admit cross-year triangles. These effects require a sequence of snapshots; one final aggregate cannot reveal when connectivity appears. The relative range is a window-sensitivity statistic, not a significance test against a temporal null, and identical density/degree sensitivity follows from the common denominator.

	PS		CM	
	real	P1	real	P1
Burstiness B	> 0	$\approx$ lower	> 0	$\approx$ lower
Memory M	> 0	≈ 0	> 0	≈ 0

Table 3. Expected qualitative outcome of M2a (values to be measured; mean $\pm$ std over ≥ 20 P1 samples).

Verdict on H1. Supported for CIT under the stated node denominator and strict-year convention. A comparative verdict for PS and CM awaits their measurements; the relative sensitivity of network types cannot yet be ranked.

5.2. Activity patterns

Burstiness and memory. [Expected] Inter-event times of both PS and CM are heavy-tailed and have positive burstiness B and memory M. Under P1 the heavy tail is largely removed, so $B_{\text{real}} - B_{\text{null}} > 0$. Contact bursts in PS are bounded by the school timetable, while CM shows longer waiting times (hours to days) and a stronger tail.

Persistence and turnover. [Expected] Many contacts in PS are repeated within the same day (within class), so the edge repetition rate is high and consecutive-snapshot Jaccard is moderate, with drops at class changes. In CM most pairs interact only a few times, so tie lifetimes are short and

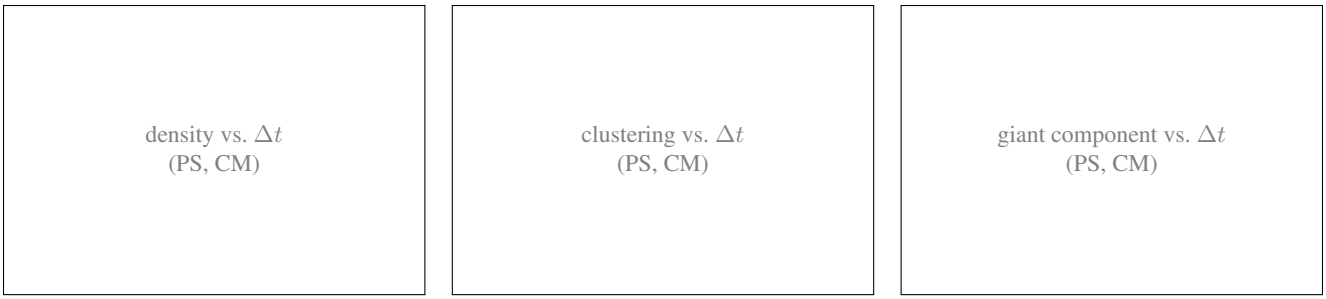


Figure 5. Structural metrics of snapshots as functions of the window length Δt; dashed lines show the static aggregate. (Placeholder.)

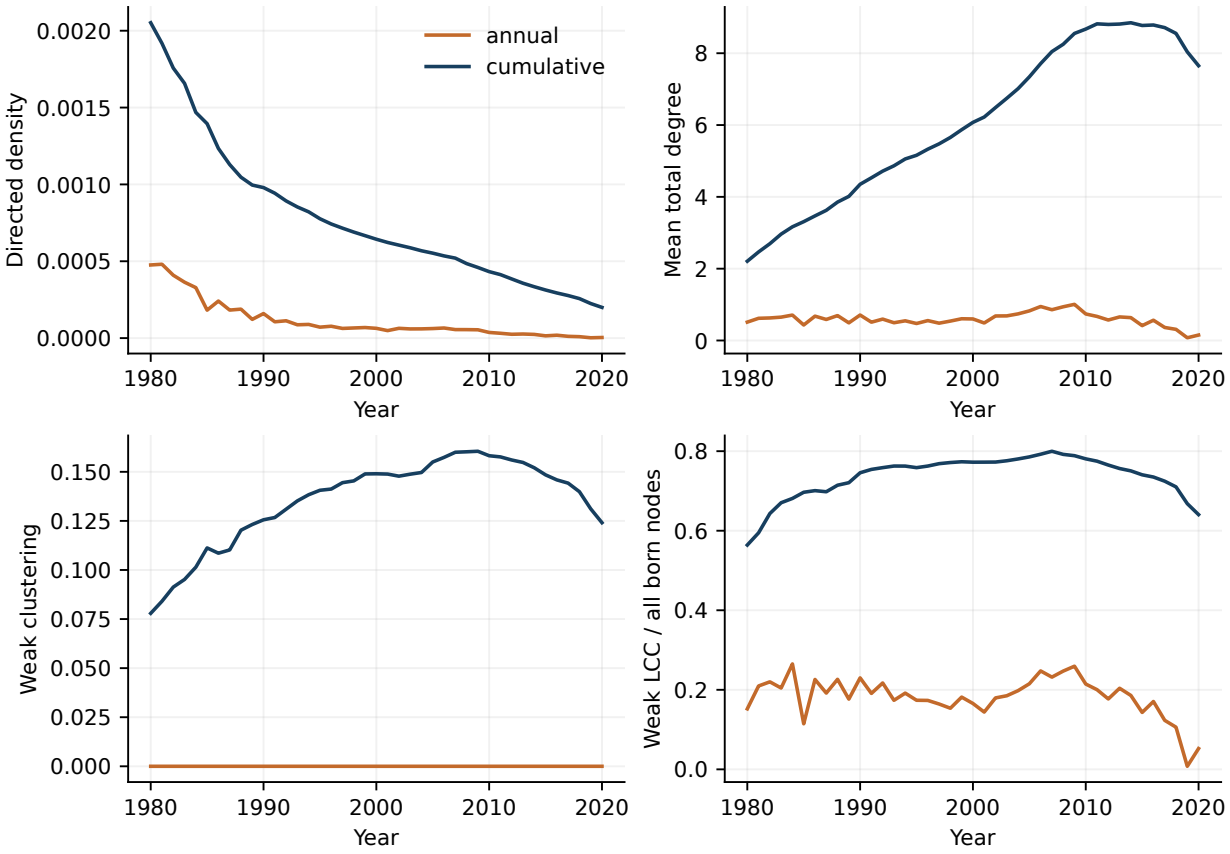


Figure 6. Measured CIT annual and cumulative structure, including born isolates. The annual zero clustering is imposed by strict publication-year ordering.

turnover is high. P1 keeps the static graph and therefore the repetition rate, but changes lifetimes and Jaccard.

Verdict on H2 and H3. *[Outline] Supported / not supported / depends on network type or window.*

[Expected] H2 supported in both. H3 supported in PS, only partly in CM.

5.3. Structural evolution

Preferential attachment (H4a). The conditional kernel $(k+1)^{\alpha}$ gives $\alpha = 0.714$, with a descriptive 95% year-bootstrap interval $[0.693, 0.730]$. Twenty P3a samples yield 0.0006 ± 0.0051 and age-only P3b yields -0.6512 ± 0.0053. The observed exponent exceeds all null values (empirical one-sided probability $1/21$ per comparison, with coarse resolution). The exponent stays positive under time splits, offsets 0.5/2 and a core venue-family subset. **Verdict:** sublin-

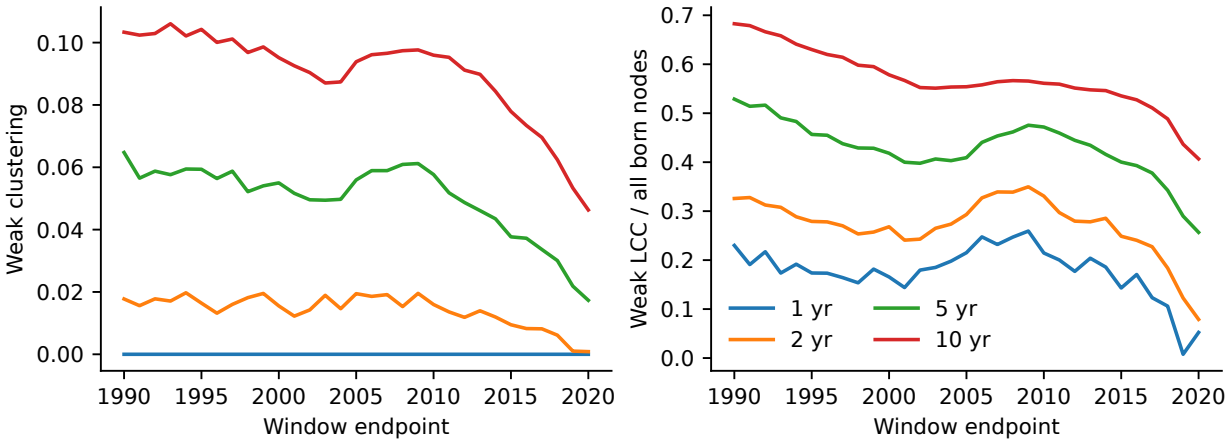


Figure 7. Measured CIT trailing-window clustering and connectivity for 1, 2, 5 and 10 years. Born isolates are included at every endpoint.

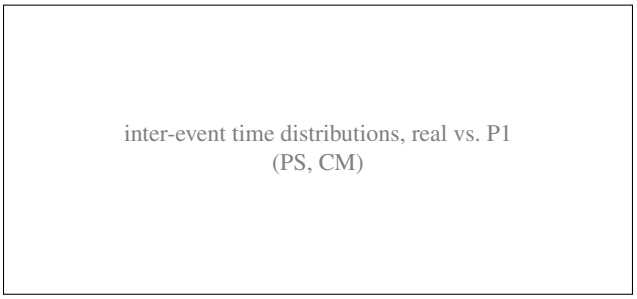


Figure 8. Inter-event time distributions on log–log axes. (Placeholder.)

ear degree preference is supported within this corpus, beyond the age-only reference; this does not identify a causal PA mechanism.

Ageing (H4b). Lognormal target preference is preferred to exponential, power-law and uniform kernels, with respective ΔAIC 537.8, 9,360.5 and 41,796.7. The lognormal parameters are $(\mu, \sigma) = (1.858, 1.108)$; its underlying scale e^{μ} is 6.41 years. Mean age over all retained citations is 5.711 years, versus 11.457 ± 0.034 for P3a and 5.717 ± 0.018 for calibrated P3b. Lognormal is preferred in all three target cohorts, although recent-cohort tails remain censored. **Verdict:** age-independent choice is inadequate; P3b's agreement is calibration, not independent confirmation of ageing.

Densification and diameter (H4c). For 1990–2020, $a = 1.343$ ($R^2 = 0.991$), with descriptive year-resampling interval $[1.288, 1.406]$. Growth is weaker in 2005–2020 ($a = 1.065$) than 1990–2004 (1.455); cutoff 2015 gives 1.427, inclusion of year ties 1.324, and the core venue subset 1.410. Densification is superlinear edge growth, not increasing directed density. The sampled weak-LCC effective diameter is 6 hops in every year; 64/256-source endpoint checks agree. P3 fixes $N(t)$ and $E(t)$, hence also a. **Verdict:** densification is observed, but diameter shrinkage is not detected at the sampled 90th-percentile resolution.

Citation dynamics and prediction (H4d). For 500 full-history fits, median WSB parameters are $(\lambda, \mu, \sigma) = (2.717, 0.834, 1.238)$, median RMSLE is 0.074 and scaled-collapse RMSE is 0.022. These are fitted-sample diagnostics. On 500 papers selected using only their first five years, age-15 MAE is 15.144 for WSB, 17.064 for linear extrapolation and 27,588 for no-ageing PA; at the observed 2020 endpoint it is 28.016, 27.271 and 2.79×10^6. The paired MAE differences WSB minus linear have 95% intervals $[-4.471, 0.806]$ and $[-4.764, 8.128]$, both including zero. Forecast fits hit parameter bounds for 38.6% of papers; all remain in primary scoring. **Verdict:** compact descriptive fits are supported, but a clear predictive advantage over the simple linear baseline is not established. Observed-endpoint citations are censored within-corpus counts, not lifetime impact.

Dynamic vs. static communities. [Expected] In PS the static partition recovers the classes well, but snapshots reveal mixing during breaks and re-formation of classroom groups during lessons, so consecutive NMI is moderate and fluctuates with the timetable. In CM snapshot communities are unstable (low consecutive NMI), and P1 lowers consecutive NMI further. Results should be insensitive to the resolution parameter within a reasonable range.

Verdict on H5. *[Outline] Supported / not supported / depends on network type or window.*

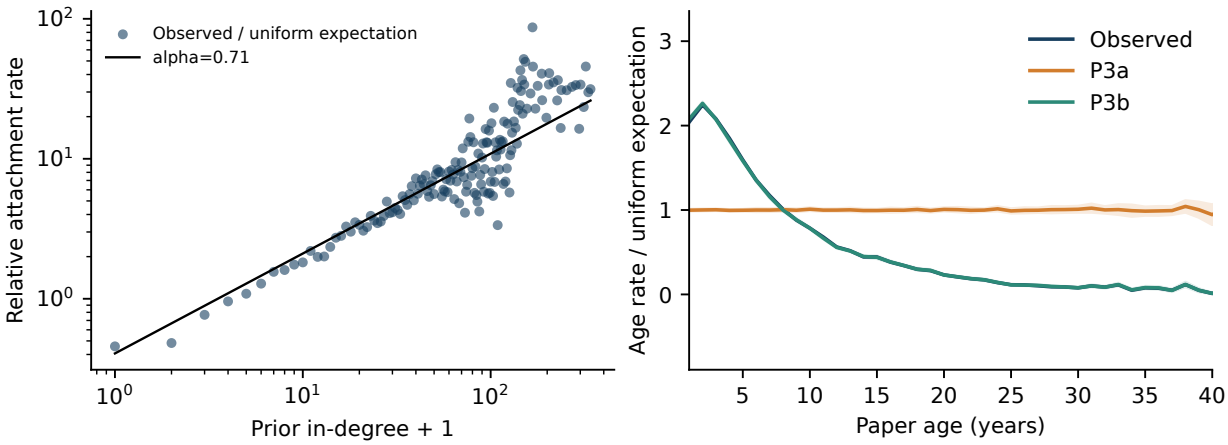


Figure 9. Measured CIT attachment and exposure-adjusted age preference. Points with at least five hits are displayed; the exponent uses all supported classes with yearly normalization. Null bands show mean ± SD.

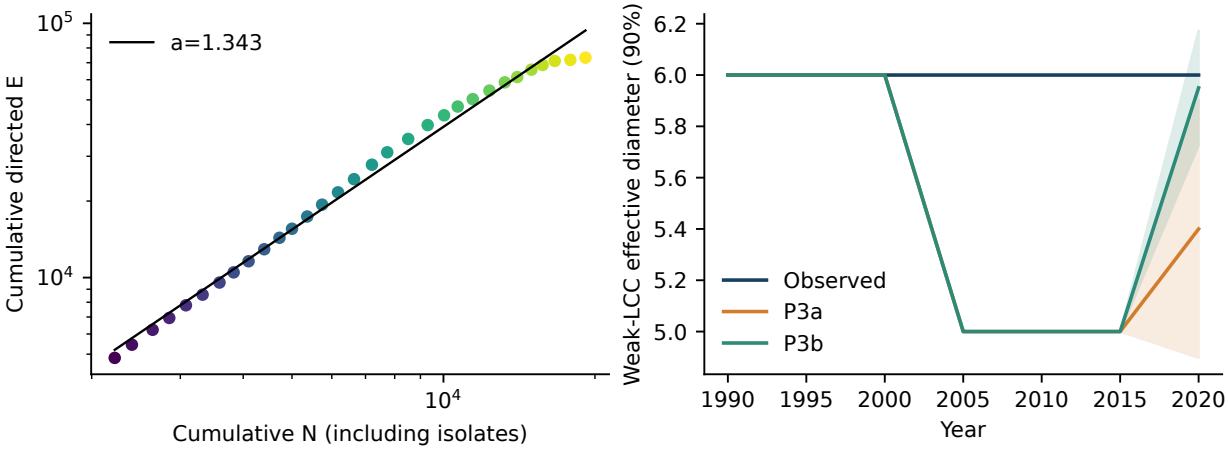


Figure 10. CIT cumulative growth and effective diameter. Null diameter means and SD are estimated every five years. P3 cannot test densification significance because its source reference counts are fixed.

	PS	CM
Consecutive NMI	medium	low
Dynamic vs. static NMI	medium	low
NMI with classes	high (static)	–

Table 4. Expected qualitative outcome of M4 (values and P1 baseline to be measured).

[Expected] H5 supported for CM, and in PS tied to the daily timetable.

5.4. Paths, centrality and spreading

Reachability. [Expected] The fraction of pairs connected by a time-respecting path is well below the static value in both networks. Under P1 reachability changes because temporal order is destroyed, whereas P2 changes it through topology, so the two nulls attribute the loss to different causes.

Centrality. [Expected] Rankings by static and temporal closeness are positively correlated but far from identical (Kendall τ clearly below 1), with the largest differences for nodes active only early or late.

Spreading. [Expected] SIR epidemics on the real sequences peak later than on P1, $T_{\text{real}}/T_{\text{null}} > 1$, and the effect is larger for the burstier CM. In PS, the final size is limited by the timetable (outbreaks pause overnight).

Verdict on H6 and H7. *[Outline] Supported / not supported / depends on network type or window.*

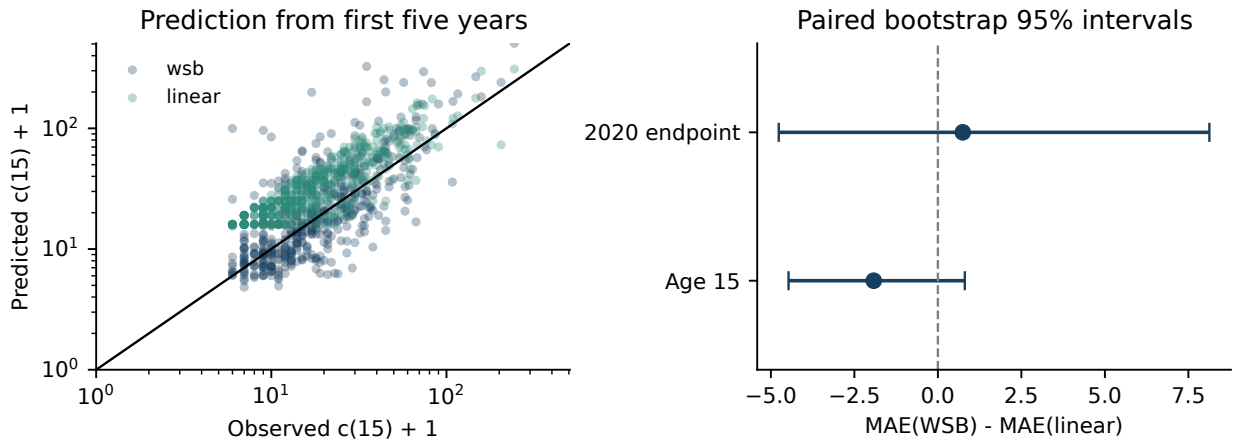


Figure 11. WSB held-out prediction and paired MAE differences versus linear. Both bootstrap intervals cross zero; fitted collapse alone does not establish long-term predictability.

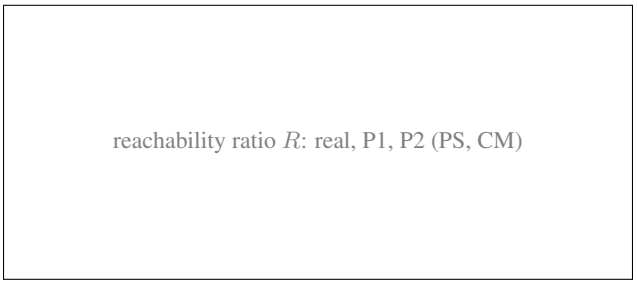


Figure 12. Temporal reachability ratio, real vs. P1 and P2. (Placeholder.)

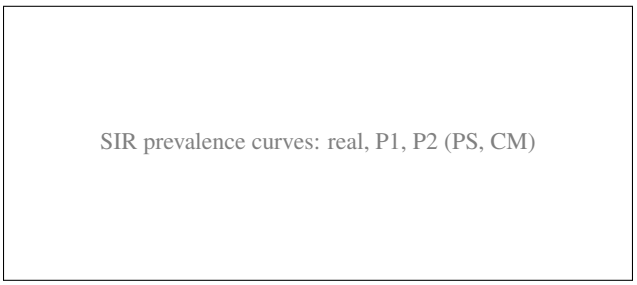


Figure 13. Mean SIR prevalence with ≥ 100 seeds. (Placeholder.)

[Expected] H6 supported in both networks. H7 supported, strongest for CM.

5.5. CIT model diagnostics and robustness

Null controls and ageing alternatives. Figure 14 visualizes the separation of the observed attachment exponent from both null ensembles and the age-kernel AIC comparison. P3b matches mean citation age because its age weights were fitted to these data; that agreement is a property of the control, not independent evidence for ageing. With only 20 realizations per null, an observed value above all nulls has empirical one-sided probability $1/21$; this does not justify a precise tail estimate. The ageing comparison is on identical risk sets and observations.

Age kernel	Parameters	ΔAIC
Lognormal	2	0.0
Exponential	1	537.8
Power-law	1	9,360.5
Uniform	0	41,796.7

Table 5. CIT age-preference alternatives on common observed support. AIC selects among the declared models, not among every possible ageing mechanism.

Fit quality does not establish predictability. The full-history sample is 500 of 1,019 papers with at least ten citations by age 15; the forecast sample is 500 of 1,220 eligible papers selected only by having at least five citations in their first five years. Both samples are drawn with a fixed seed. Median full-history RMSLE is 0.0736 and median scaled-collapse RMSE 0.0220 (90th percentile 0.0453); 10.2% of full-history fits hit a parameter bound. Figure 15 is an in-sample diagnostic. Held-out scoring in Tab. 6 and Fig. 11 provides the separate predictive test. In particular, a collapse obtained using parameters fitted to the same curve is not independent evidence for universal citation dynamics.

Sensitivity of the conclusions. Table 7 reports time splits, cutoff and corpus-selection checks. The sign of degree preference and superlinear full-period growth persist, but a is substantially weaker in the late period. Degree offsets 0.5 and 2 give $\alpha = 0.604$ and 0.858, so the numeri-

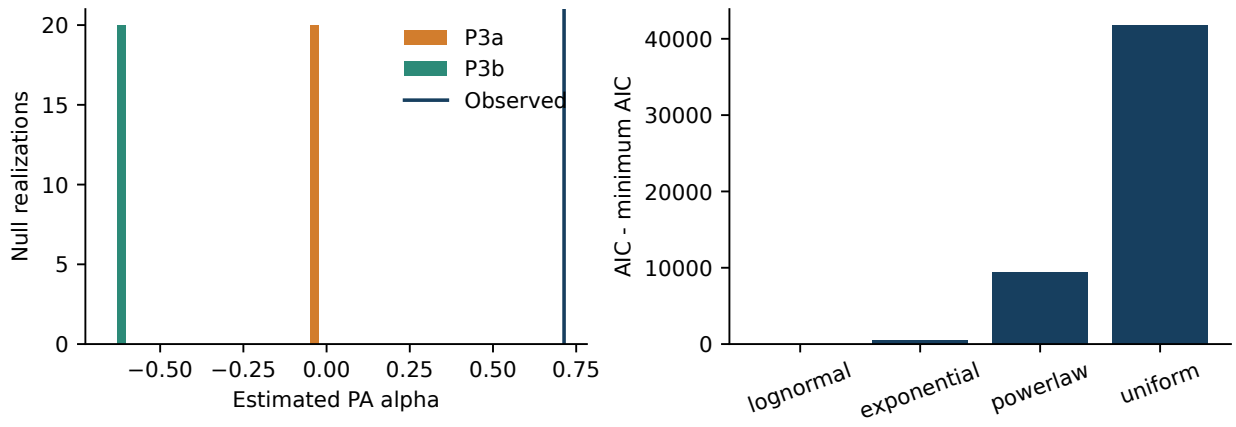


Figure 14. CIT null-model attachment exponents and age-kernel AIC differences. Uniform P3a and calibrated age-only P3b each have 20 independent realizations.

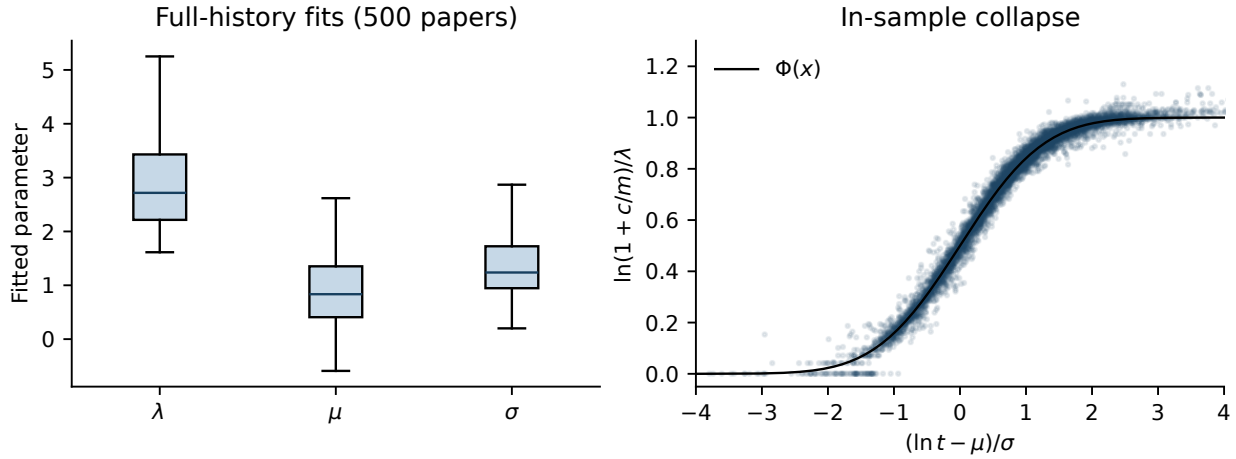


Figure 15. WSB full-history diagnostics: parameter distributions (boxes are interquartile ranges, median lines, standard 1.5-IQR whiskers; outliers hidden) and the scaled citation histories against $\Phi(x)$. Parameters are fitted to ages 1–15, so the collapse is descriptive, not held-out validation.

Target	Model	MAE	RMSLE
15 yr	WSB	15.144	0.601
	Linear	17.064	0.662
	No-age PA	27,588	4.601
2020	WSB	28.016	0.778
	Linear	27.271	0.876
	No-age PA	2.79×10^{6}	7.000

Table 6. Matched forecasts on 500 selected papers using only years 1–5 for training. The 2020 target is the observed, censored corpus endpoint. The exponential no-age PA baseline overextrapolates; beating it alone is a weak test. The paired MAE intervals against linear both include zero.

cal exponent is convention-dependent. Including 5,486 year ties gives $a = 1.324$; ties remain excluded from the primary DAG. Effective-diameter endpoint estimates remain 6 hops with 64, 128 or 256 sources. First-ten versus all-twenty null means of α are 0.00055/0.00059 (P3a) and $-0.65320/-0.65118$ (P3b). Lognormal ageing is preferred in all target cohorts and both observation halves, but the recent cohort's fitted tail extends beyond its observed follow-up. These checks support qualitative stability without eliminating coverage bias, right censoring or unobserved fitness.

5.6. Measured hypothesis summary

Table 8 separates available CIT evidence from pending PS/CM experiments. No ordering of network types by temporal sensitivity is established until the same normalized

Variant	α	a
1990–2020 (primary)	0.714	1.343
1990–2004	0.684	1.455
2005–2020	0.721	1.065
Cutoff 2015	0.722	1.427
Core venue families	0.731	1.410
Same-year edges included	–	1.324

Table 7. CIT robustness from the saved run. Core venue families contain 14,235 nodes and 47,009 primary edges; the year-tie variant is structural only.

ID	PS	CM	Measured CIT evidence
H1	Pending	Pending	Relative ranges: density/degree 1.798, clustering 2.361, LCC 1.114.
H2	Pending	Pending	–
H3	Pending	Pending	–
H4a	–	–	$\alpha = 0.714$ [0.693, 0.730]; sublinear degree preference.
H4b	–	–	Lognormal preferred; scale 6.41 yr; ΔAIC versus exponential 537.8.
H4c	–	–	$a = 1.343$; $D_{90} = 6$ throughout. Densification, no detected shrinkage.
H4d	–	–	Compact fits; MAE advantage over linear unresolved.
H5	Pending	Pending	–
H6	Pending	Pending	–
H7	Pending	Pending	–

Table 8. Available evidence. Pending denotes unmeasured PS and CM experiments; – denotes inapplicability. CIT intervals are descriptive year resamples. H4c/H4d have mixed verdicts. Values come from the shared summary and supporting tables.

measures have been computed for all applicable datasets. The split H4 verdict avoids treating densification, diameter shrinkage, descriptive fit and predictive superiority as one interchangeable claim.

6. Discussion

What the static network already tells us. Conclusions that survive aggregation can be read from Tab. 8. For CIT, the final aggregate describes degree heterogeneity, clustering and connected components. It cannot by itself identify preferential attachment or ageing: these compare new citations with prior degrees and paper ages. Even densification requires a sequence of historical $N(t), E(t)$ pairs, rather than one final graph. The PS/CM claims remain to be tested.

What requires time. Some conclusions need windows (Parts I and III) and others the full event order (Parts II and IV). [Expected] Burstiness, turnover, community reorganisation, time-respecting reachability and spreading speed cannot be obtained from the aggregate.

Why network types differ. The three networks differ in generating mechanism (physical proximity, voluntary messaging, citing earlier work), time scale and edge semantics (a contact ends, a message is a point event, a citation persists). [Expected] Proximity is constrained by timetables and space and thus has strong periodic structure, messages are driven by individual bursty decisions. CIT has persistent edges and measured age-dependent target preferences. A cross-network ordering still needs PS/CM measurements.

Robustness. We check the choice of window size, the observation period (first vs. second half), the community-detection resolution and the choice of null model. For CIT, the full-period densification exponent is 1.343 but falls to 1.065 in the later half, while remaining 1.427 with cutoff 2015. Positive degree preference persists under all declared variants, although its exponent depends on the degree offset. The diameter verdict is unchanged by the sampled-source sensitivity checks. These are qualitative robustness findings, not evidence that coverage and censoring have been removed.

Citation-specific evidence. CIT supports degree preference and age-dependent target choice, but separates two claims often grouped together: cumulative edges grow superlinearly, while the sampled effective diameter does not shrink. A three-parameter citation curve fits selected histories well, yet its paired MAE advantage over linear prediction is unresolved. Annual zero clustering is imposed by the strict-year snapshot construction. These examples show why representation, fitted-sample description and held-out prediction require separate interpretations. Late reference coverage, within-venue selection and right censoring constrain all citation conclusions.

Limitations. There is one dataset per network type, so generality is limited and conclusions hold "in these data". CIT also has within-venue sampling, annual timestamp ties and changing reference coverage. Its year-block bootstrap intervals are descriptive; cumulative graph observations are dependent. The 20-draw null ensembles give coarse tail resolution. Forecast selection excludes low-cited papers, and 38.6% of short-history WSB fits hit parameter bounds; fitted collapse is not independent validation. Boundary effects and sampling bias remain (finite two-day contact window, sensor participation). Citation networks are degenerate temporal networks (DAG, no deletion), which is why several

methods do not apply. Each system also contains several time scales [10], so a window that suits one process may hide another. Finally, SocioPatterns data are released under CC BY-NC-SA.

7. Conclusion

The completed citation-network experiments show that the importance of time depends on the question. A final aggregate reveals degree heterogeneity and connectivity, while preferential attachment, ageing and the growth law require historical degrees, publication ages or cumulative snapshots. In the selected AMiner database corpus, sublinear degree preference and age-dependent citation choice coexist with superlinear edge growth, yet the sampled effective diameter does not shrink. WSB compactly describes selected citation histories, but its mean-absolute-error advantage over a simple linear forecast is not established. Thus descriptive fit, temporal mechanism and predictive value should be assessed separately. PS/CM experiments remain pending, so conclusions about their dynamics and a cross-network ranking of temporal sensitivity await measured evidence. Future work includes broader corpus coverage, more datasets per type, finer-resolution citation dates and predictive tests under alternative censoring and missing-reference assumptions.

References

[1] Albert-László Barabási. The origin of bursts and heavy tails in human dynamics. *Nature*, 435:207–211, 2005.

[2] Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999.

[3] Vincent D. Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, page P10008, 2008.

[4] S. N. Dorogovtsev and J. F. F. Mendes. Evolution of networks with aging of sites. *Physical Review E*, 62:1842–1845, 2000.

[5] Laetitia Gauvin, Mathieu Génois, Márton Karsai, Mikko Kivelä, Taro Takaguchi, Eugenio Valdano, and Christian L. Vestergaard. Randomized reference models for temporal networks. *SIAM Review*, 64(4):763–830, 2022.

[6] Valerio Gemmetto, Alain Barrat, and Ciro Cattuto. Mitigation of infectious disease at school: targeted class closure vs school closure. *BMC Infectious Diseases*, 14:695, 2014.

[7] K.-I. Goh and A.-L. Barabási. Burstiness and memory in complex systems. *EPL (Europhysics Letters)*, 81:48002, 2008.

[8] Derek Greene, Donal Doyle, and Padraig Cunningham. Tracking the evolution of communities in dynamic social networks. In *International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pages 176–183, 2010.

[9] Petter Holme. Network reachability of real-world contact sequences. *Physical Review E*, 71:046119, 2005.

[10] Petter Holme. Modern temporal network theory: a colloquium. *The European Physical Journal B*, 88:234, 2015.

[11] Petter Holme and Jari Saramäki. Temporal networks. *Physics Reports*, 519(3):97–125, 2012.

[12] H. Jeong, Z. Néda, and A.-L. Barabási. Measuring preferential attachment in evolving networks. *EPL (Europhysics Letters)*, 61(4):567–572, 2003.

[13] M. Karsai, M. Kivelä, R. K. Pan, K. Kaski, J. Kertész, A.-L. Barabási, and J. Saramäki. Small but slow world: How network topology and burstiness slow down spreading. *Physical Review E*, 83:025102, 2011.

[14] Hyoungshick Kim and Ross Anderson. Temporal node centrality in complex networks. *Physical Review E*, 85:026107, 2012.

[15] Lauri Kovanen, Márton Karsai, Kimmo Kaski, János Kertész, and Jari Saramäki. Temporal motifs in time-dependent networks. *Journal of Statistical Mechanics: Theory and Experiment*, page P11005, 2011.

[16] Gautier Krings, Márton Karsai, Sebastian Bernhardsson, Vincent D. Blondel, and Jari Saramäki. Effects of time window size and placement on the structure of an aggregated communication network. *EPJ Data Science*, 1:4, 2012.

[17] Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. Graphs over time: densification laws, shrinking diameters and possible explanations. In *Proceedings of the 11th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 177–187, 2005.

[18] Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. Graph evolution: densification and shrinking diameters. *ACM Transactions on Knowledge Discovery from Data*, 1(1):2, 2007.

[19] Naoki Masuda and Renaud Lambiotte. *A Guide to Temporal Networks*. World Scientific, 2016.

[20] Cristopher Moore, Gourab Ghoshal, and M. E. J. Newman. Exact solutions for models of evolving networks with addition and deletion of nodes. *Physical Review E*, 74:036121, 2006.

[21] Peter J. Mucha, Thomas Richardson, Kevin Macon, Mason A. Porter, and Jukka-Pekka Onnela. Community structure in time-dependent, multiscale, and multiplex networks. *Science*, 328(5980):876–878, 2010.

[22] M. E. J. Newman. Clustering and preferential attachment in growing networks. *Physical Review E*, 64:025102, 2001.

[23] Raj Kumar Pan and Jari Saramäki. Path lengths, correlations, and centrality in temporal networks. *Physical Review E*, 84: 016105, 2011.

[24] Pietro Panzarasa, Tore Opsahl, and Kathleen M. Carley. Patterns and dynamics of users' behavior and interaction: Network analysis of an online community. *Journal of the American Society for Information Science and Technology*, 60(5): 911–932, 2009.

[25] Ashwin Paranjape, Austin R. Benson, and Jure Leskovec. Motifs in temporal networks. In *Proceedings of the Tenth ACM International Conference on Web Search and Data Mining (WSDM)*, pages 601–610, 2017.

[26] Nicola Perra, Bruno Gonçalves, Romualdo Pastor-Satorras, and Alessandro Vespignani. Activity driven modeling of time varying networks. *Scientific Reports*, 2:469, 2012.

[27] Bruno Ribeiro, Nicola Perra, and Andrea Baronchelli. Quantifying the effect of temporal resolution on time-varying networks. *Scientific Reports*, 3:3006, 2013.

[28] Giulio Rossetti and Rémy Cazabet. Community discovery in dynamic networks: a survey. *ACM Computing Surveys*, 51 (2):35, 2018.

[29] Martin Rosvall and Carl T. Bergstrom. Mapping change in large networks. *PLoS ONE*, 5(1):e8694, 2010.

[30] Jari Saramäki, E. A. Leicht, Eduardo López, Sam G. B. Roberts, Felix Reed-Tsochas, and Robin I. M. Dunbar. Persistence of social signatures in human communication. *Proceedings of the National Academy of Sciences*, 111(3):942–947, 2014.

[31] Ben Steer, Naomi Arnold, Cheick Tidiane Ba, Renaud Lambiotte, et al. Raphtory: The temporal graph engine for rust and python. *Journal of Open Source Software*, 9(95):5940, 2024.

[32] Juliette Stehlé, Nicolas Voirin, Alain Barrat, Ciro Cattuto, Lorenzo Isella, Jean-François Pinton, Marco Quaggiotto, Wouter Van den Broeck, Corinne Régis, Bruno Lina, and Philippe Vanhems. High-resolution measurements of face-to-face contact patterns in a primary school. *PLoS ONE*, 6 (8):e23176, 2011.

[33] Jie Tang, Jing Zhang, Limin Yao, Juanzi Li, Li Zhang, and Zhong Su. Arnetminer: extraction and mining of academic social networks. In *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 990–998, 2008.

[34] V. A. Traag, L. Waltman, and N. J. van Eck. From louvain to leiden: guaranteeing well-connected communities. *Scientific Reports*, 9:5233, 2019.

[35] Dashun Wang, Chaoming Song, and Albert-László Barabási. Quantifying long-term scientific impact. *Science*, 342(6154): 127–132, 2013.